Write short notes:

a) Energetics of Glycolysis

Glycolysis is the metabolic pathway that breaks down **glucose ($C_6H_{12}O_6$)** into **2 molecules of pyruvate**, producing **ATP and NADH** in the process. It occurs in the **cytoplasm** and does **not require oxygen** (anaerobic).

Net Reaction of Glycolysis:

Glucose + 2 NAD^+ + 2 ADP + 2 Pi → 2 Pyruvate + 2 NADH + 2 ATP + 2 H_2O + 2 H^+

Energy Investment Phase (Steps 1–5):

- **2 ATP used**

Energy Payoff Phase (Steps 6–10):

- **4 ATP produced**
- **2 NADH produced**

Net Energy Gain:

- **ATP: 4 produced – 2 used = 2 net ATP**
- **NADH: 2 molecules (each NADH yields about 2.5 ATP in aerobic respiration)**

Summarize that:

- **Net ATP gain = 2**
- **Net NADH gain = 2**
- Occurs in cytoplasm, anaerobically

Energetics of Glycolysis (1)
Four ATP molecules are produced in the payoff phase whereas 2 ATP molecules are consumed during the preparatory phase. Two molecules of NADH are formed in the pay-off phase. Thus the net yield in the process of glycolysis is 2 ATP and 2 NADH molecules.

Explain pH

pH is a fundamental concept in chemistry and biology that helps us understand the acidic or basic nature of substances. It is crucial in maintaining proper conditions for life processes, industrial applications, and environmental monitoring.

pH is the **measure of hydrogen ion concentration** $[H^{+}]$ in a solution. It indicates how many hydrogen ions are present, which determines the solution's acidity or basicity.**pH** stands for **"potential of hydrogen"** or **"power of hydrogen"** and is a scale used to measure how acidic or basic (alkaline) a solution is.

$$pH = -\log_{10}[H^{+}]$$

- If [H+] is high → pH is **low** → solution is **acidic**
- If [H+] is low → pH is **high** → solution is **basic (alkaline)**

pH Scale:

- The pH scale ranges from **0 to 14**:

pH Value	Nature of Solution
0 – 6.9	Acidic
7	Neutral
7.1 – 14	Basic (Alkaline)

- **Neutral solution** (like pure water) has a pH of **7**
- **Stomach acid** has a pH around **1–2** (very acidic).
- **Soap or bleach** has a pH around **9–13** (basic).

Importance of pH:

1. **Biological systems:**
   - Enzymes work best at specific pH levels (e.g., pepsin in the stomach at pH ~2).
   - Blood must maintain a pH of **7.35–7.45** for normal physiological functions.
2. **Agriculture:**
   - Soil pH affects nutrient availability for plants.
3. **Industry:**
   - pH control is essential in processes like fermentation, water treatment, and cosmetics production.
4. **Environmental science:**
   - pH of water bodies affects aquatic life and environmental heal

Define and discuss the term Bio-chemistry. Explain life in terms of biochemistry with example. Discuss your role as a biochemist in the field of agriculture	Discuss the importance of biological buffer system.

Ans:

Life in Terms of Biochemistry (with Example)

Life can be understood at the molecular level through biochemical processes. All living organisms—from simple bacteria to complex humans—carry out a series of chemical reactions that are tightly regulated to maintain homeostasis.

Example:
Cellular respiration is a fundamental biochemical process that illustrates life in action. In this process, glucose ($C_6H_{12}O_6$), a carbohydrate, is broken down in the presence of oxygen to release energy in the form of **ATP (adenosine triphosphate)**, which is used for various cellular functions.

Equation:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy(ATP)$$

This process shows how biochemistry explains the transformation of chemical energy into usable biological energy—essential for life.

Role of a Biochemist in the Field of Agriculture

As a biochemist in agriculture, your role is crucial in enhancing crop yield, improving soil health, and developing sustainable farming practices. Some key contributions include:

1. **Crop Improvement:**
   - Study plant biochemistry to develop genetically modified crops that are more resistant to pests, diseases, and environmental stresses like drought and salinity.
2. **Fertilizer and Pesticide Development:**
   - Design eco-friendly fertilizers and biopesticides by understanding the biochemical needs of plants and the life cycles of pests.
3. **Soil Biochemistry:**
   - Analyze soil enzymes and microbial activity to evaluate soil health and nutrient availability.
4. **Plant Nutrition:**
   - Monitor biochemical markers in plants to assess nutrient deficiencies and recommend appropriate supplements.
5. **Post-Harvest Technology:**
   - Study the biochemical changes in fruits and vegetables after harvest to enhance shelf-life and reduce spoilage.

Conclusion

Biochemistry is fundamental to understanding life at the molecular level. By studying the biochemical processes that sustain organisms, we gain insight into health, disease, and the environment. In agriculture, a biochemist plays a pivotal role in driving innovation and ensuring food security through scientific solutions.

Definition and Discussion of Biochemistry

Biochemistry is the branch of science that explores the chemical processes and substances that occur within living organisms. It is an interdisciplinary science that combines principles of biology and chemistry to study the molecular mechanisms of life.

It focuses on the structure, composition, and chemical reactions of biomolecules such as carbohydrates, proteins, lipids, nucleic acids, and enzymes. Biochemistry seeks to understand how these molecules interact to sustain life processes like metabolism, growth, reproduction, and heredity.

Importance of Biological Buffer Systems

A **biological buffer system** is a solution that resists changes in pH when small amounts of acids or bases are added. These systems are essential in maintaining a stable internal environment (**homeostasis**) in living organisms.

Why Biological Buffers Are Important:

1. Maintain pH for Enzyme Activity

- Most enzymes function within a narrow pH range.
- Even slight changes in pH can denature enzymes or alter their activity.
- Buffers help maintain the optimal pH needed for biochemical reactions to occur efficiently.

2. Support Metabolic Processes

- Cellular respiration, protein synthesis, and other metabolic activities depend on constant pH.
- Buffers stabilize the pH during these processes, ensuring normal cellular function.

3. Ensure Proper Blood Function

- The **blood buffer system** (mainly **bicarbonate buffer**) keeps blood pH between **7.35 and 7.45**.
- If blood becomes too acidic or too alkaline, it can lead to life-threatening conditions (acidosis or alkalosis).

4. Maintain Cellular Environment

- Intracellular buffers (like **phosphate buffers**) maintain pH within cells.
- This is crucial for DNA replication, protein synthesis, and ion transport across membranes.

5. Prevent Damage from pH Fluctuations

- Sudden pH changes can damage tissues and interfere with organ function.
- Buffers provide resistance to such changes, especially during intense exercise or stress.

Examples of Biological Buffers

Buffer System	Location	Function
Bicarbonate buffer	Blood plasma	Regulates blood pH
Phosphate buffer	Cells and kidneys	Controls intracellular pH
Protein buffers (e.g., hemoglobin)	Red blood cells	Buffering capacity in blood
Ammonia buffer	Kidneys	Helps excrete H^+ ions in urine

How do you classify enzyme?

Ans: Classification of Enzymes: All the enzymes are broadly classified into 6 major classes:

1. Oxidoreductases
2. Transferases
3. Hydrolases
4. Lyases
5. Isomerases
6. Ligases

Enzymes are classified based on the type of chemical reaction they catalyze. The International Union of Biochemistry and Molecular Biology (IUBMB) classifies enzymes into six major classes, each with subclasses. These are:

1. Oxidoreductases
   - Catalyze oxidation-reduction reactions
   - Example: Dehydrogenase, Oxidase
2. Transferases
   - Transfer a functional group (e.g., methyl, phosphate) from one molecule to another
   - Example: Transaminase, Kinase
3. Hydrolases
   - Catalyze hydrolysis reactions (breaking bonds using water)
   - Example: Protease, Lipase, Amylase
4. Lyases
   - Add or remove groups to form double bonds (not hydrolysis or oxidation)
   - Example: Decarboxylase, Aldolase
5. Isomerases
   - Catalyze isomerization (rearrangement within a molecule)
   - Example: Racemase, Epimerase
6. Ligases (or Synthetases)
   - Join two molecules using ATP or another energy source
   - Example: DNA ligase, Synthetase

Each enzyme is also given an EC (Enzyme Commission) number for specific identification, like EC 1.1.1.1 for alcohol dehydrogenase.

Question: What are the factors affecting enzyme activity? Briefly explain any one.

Ans: Factors Affecting Enzyme activities are as following:

Tempre
pH level
Substrate Concentration
Enzyme Concentration
Presence of inhibitor (Competitive or non competitive)
Cofactor and Coenzyme
Allosteric regulations and Product Concentration

pH level: Enzymes have an optimal pH range within which they function best. This range is determined by the amino acid residues in the enzyme's active site and their ability to interact with the substrate. If the pH deviates from this optimal range, the enzyme's structure can be altered, leading to a decrease or even loss of activity. This is because changes in pH can affect the ionization state of the amino acid residues, disrupting the hydrogen bonds and ionic interactions that maintain the enzyme's shape. For example, if the pH is too acidic or alkaline, the enzyme may become denatured, meaning its shape is changed and it can no longer bind to the substrate.

Question : Differentiate between essential and non-essential amino acids.

Ans: **Difference Between Essential and Non-Essential Amino Acids**

Amino acids are the building blocks of proteins. They are categorized into **essential** and **non-essential** based on whether the human body can synthesize them.

Essential amino acids: Must be taken from food.

Non-essential amino acids: Body can make them.

Feature	Essential Amino Acids	Non-Essential Amino Acids
Definition	Cannot be synthesized by the body and must be obtained from diet	Can be synthesized by the body from other compounds
Source	Must be obtained through food (e.g., meat, eggs, dairy, legumes)	Body produces them internally
Examples	Leucine, Isoleucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine, Histidine*	Alanine, Asparagine, Aspartic acid, Glutamic acid, Serine, Tyrosine, Cysteine, Glutamine, Glycine, Proline
Nutritional Importance	Critical for growth, tissue repair, and nutrient absorption	Important but not diet-dependent for healthy individuals
Deficiency Effects	Can lead to muscle loss, weak immunity, fatigue, poor growth	Usually no deficiency unless in cases of metabolic disorders

*Histidine is essential for infants and growing children.**:**

Question: What do you know about chemiosmotic theory?

Ans: The chemiosmotic theory was proposed by British biochemist Peter Mitchell in 1961 to explain how cells synthesize ATP (adenosine triphosphate)—the primary energy currency of life—during processes like cellular respiration and photosynthesis. This theory revolutionized our understanding of bioenergetics and earned Mitchell the Nobel Prize in Chemistry in 1978.

The chemiosmotic theory describes a process where:

1. Electron Transport: Electrons are transferred through a series of proteins known as the electron transport chain (ETC), located in the inner mitochondrial membrane (in respiration) or the thylakoid membrane (in photosynthesis).
2. Proton Pumping: The energy released from electron transfers is used to pump protons (H^+ ions) across the membrane, creating a proton gradient—a difference in proton concentration and electric charge across the membrane.

3. Proton-Motive Force: This gradient generates a proton-motive force (PMF), which stores potential energy due to the difference in proton concentration and charge.
4. ATP Synthesis: Protons flow back across the membrane through the enzyme ATP synthase, driven by the PMF. This flow provides the energy needed to convert ADP (adenosine diphosphate) and inorganic phosphate (Pi) into ATP.

It is occurs in different ways in-

- Mitochondria: During oxidative phosphorylation, part of cellular respiration, the ETC creates a proton gradient across the inner mitochondrial membrane, leading to ATP production.
- Chloroplasts: In photophosphorylation, during photosynthesis, light energy drives the ETC in the thylakoid membrane, creating a proton gradient used to synthesize ATP.
- Prokaryotes: Bacteria and archaea utilize similar mechanisms across their plasma membranes for ATP generation.

🔑 Key Points

- ATP Synthase: A crucial enzyme that synthesizes ATP using the energy from the proton gradient.
- Proton Gradient: The difference in proton concentration across a membrane, essential for ATP production.
- Electron Transport Chain: A series of protein complexes that transfer electrons and pump protons to create the gradient.

Difference between following:

(a) Amylose and amylopecti

(b) Transamination and deamination

(c) Ribonucleic acid and deoxyribonucleic acid

Ans:

(a) Amylose vs Amylopectin

Feature	Amylose	Amylopectin
Structure	Linear/unbranched chain of glucose units	Branched chain of glucose units
Type of bond	α(1→4) glycosidic bonds	α(1→4) and α(1→6) glycosidic bonds
Solubility in water	Less soluble	More soluble
Appearance	Forms a helical structure	Has a tree-like, branched structure
Proportion in starch	~20–30% of natural starch	~70–80% of natural starch

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(b) Transamination vs Deamination

Feature	Transamination	Deamination
Definition	Transfer of an amino group from one amino acid to a keto acid	Removal of an amino group from an amino acid
Main purpose	To form new (non-essential) amino acids	To break down amino acids and release nitrogen
By-products	A new amino acid and a new keto acid	Ammonia (NH_3) or ammonium ion (NH_4^+), and keto acid
Enzymes involved	Transaminases (aminotransferases)	Deaminases
Occurs mainly in	Liver and muscle	Liver

(c) RNA (Ribonucleic Acid) vs DNA (Deoxyribonucleic Acid)

Feature	RNA	DNA
Sugar	Ribose	Deoxyribose (lacks one oxygen atom at 2' carbon)
Strands	Usually single-stranded	Double-stranded helix
Bases	A, U, C, G (Uracil instead of Thymine)	A, T, C, G
Location	Cytoplasm and nucleus	Mainly in the nucleus
Stability	Less stable (more reactive due to OH group)	More stable
Function	Protein synthesis (mRNA, tRNA, rRNA)	Genetic information storage

(a) Starch vs Glycogen

Feature	Starch	Glycogen
Found in	Plants	Animals and humans
Structure	Made of amylose (linear) and amylopectin (branched)	Highly branched polymer of glucose
Degree of branching	Less branched (especially amylose)	More extensively branched
Storage location	Stored in plant cells (e.g., seeds, tubers)	Stored in liver and muscles
Function	Energy storage in plants	Energy storage in animals
Solubility	Less soluble	More soluble in water

(b) Saturated vs Unsaturated Fatty Acid

Feature	Saturated Fatty Acid	Unsaturated Fatty Acid
Chemical bonds	No double bonds between carbon atoms	One or more double bonds present
Structure	Straight chain	Bent or kinked at double bond(s)
Physical state (room temp)	Solid (e.g., butter, lard)	Liquid (e.g., vegetable oil, olive oil)
Health effects (excess)	May increase LDL (bad cholesterol)	Can help reduce LDL and increase HDL
Source	Animal fats, dairy	Plant oils, nuts, fish

(c) Transcription vs Translation

Feature	Transcription	Translation
Definition	Process of making RNA from DNA	Process of making protein from mRNA
Occurs in	Nucleus (in eukaryotes)	Cytoplasm (at ribosomes)
Product	mRNA, tRNA, or rRNA	Polypeptide (protein)
Enzyme involved	RNA polymerase	Ribosomes (and tRNA for amino acid delivery)
Template used	DNA	mRNA
Base pairing	A-U, T-A, G-C, C-G	Codons (mRNA) matched to anticodons (tRNA)

(a) Difference between Acid and Base

Property	Acid	Base
Taste	Sour	Bitter
pH Range	Less than 7	Greater than 7
Ion Produced in Water	Produces H^+ ions	Produces OH^- ions
Litmus Test	Turns blue litmus red	Turns red litmus blue
Example	Hydrochloric acid (HCl), Acetic acid (CH_3COOH)	Sodium hydroxide (NaOH), Ammonia (NH_3)

(b) Difference between Reducing Sugar and Non-reducing Sugar

Property	Reducing Sugar	Non-reducing Sugar
Reactivity	Can reduce Fehling's or Benedict's solution	Cannot reduce Fehling's or Benedict's solution
Free Functional Group	Has a free aldehyde or ketone group	No free aldehyde or ketone group
Examples	Glucose, Fructose, Maltose, Lactose	Sucrose, Trehalose
Detection	Gives positive test with Benedict's reagent	Gives negative test with Benedict's reagent

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(c) Difference between Glycosidic Bond and Peptide Bond

Property	Glycosidic Bond	Peptide Bond
Type of Molecule	Carbohydrates	Proteins (Amino acids)
Function	Joins two monosaccharides	Joins two amino acids
Chemical Nature	Ether linkage (C–O–C)	Amide linkage (–CO–NH–)
Formation	Formed by condensation between two sugar molecules	Formed by condensation between amino and carboxyl groups
Example	Maltose (glucose–glucose bond)	Dipeptides like alanine–glycine

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Define protein and classify them with examples.

Ans:

Definition of Protein:

Proteins are large, complex biomolecules made up of **amino acids** linked together by **peptide bonds**. They are essential for the structure, function, and regulation of the body's tissues and organs. Proteins perform various roles such as **catalyzing metabolic reactions (enzymes)**, **transporting molecules**, **supporting immune functions**, and **building cell structures**.

Classification of Proteins (with Examples):

Proteins can be classified based on:

1. Based on Structure:

Type	Description	Examples
Fibrous Proteins	Long, thread-like shape; structural role; insoluble in water	Collagen, Keratin, Elastin
Globular Proteins	Spherical shape; functional role; soluble in water	Hemoglobin, Enzymes, Antibodies

2. Based on Composition:

Type	Description	Examples
Simple Proteins	Made of only amino acids	Albumin, Globulin
Conjugated Proteins	Protein + non-protein part (prosthetic group)	Hemoglobin (with heme), Lipoproteins
Derived Proteins	Breakdown products of simple or conjugated proteins	Peptones, Proteoses

3. Based on Function

Type of Protein	Function	Examples
Enzymatic	Catalyze biochemical reactions	Amylase, Lipase
Structural	Provide support and shape	Collagen, Keratin
Transport	Carry substances across membranes or in blood	Hemoglobin, Myoglobin
Storage	Store amino acids or ions	Ferritin (iron storage)
Defensive	Protection against disease	Antibodies (immunoglobulins)
Hormonal	Regulate physiological processes	Insulin, Growth hormone

Define Lipid with its types and write it's importances

Ans: compounds that are **insoluble in water** but **soluble in organic solvents** like alcohol, ether, and chloroform. They are made mainly of **carbon (C), hydrogen (H), and oxygen (O)** and serve as a major source of **energy**, **cell membrane components**, and **hormone precursors**.

Types of Lipids:

Lipids are broadly classified into the following categories:

1. Simple Lipids

- Made of fatty acids and alcohol.
- **Examples:**
  - **Fats and oils** (triglycerides)
  - **Waxes** (fatty acid + long-chain alcohol)

2. Compound (Conjugated) Lipids

- Contain fatty acids, alcohol, and other chemical groups like phosphate or sugar.
- **Examples:**
  - **Phospholipids** (lipids + phosphate group) – e.g., Lecithin (found in cell membranes)
  - **Glycolipids** (lipids + carbohydrates)
  - **Lipoproteins** (lipids + proteins)

3. Derived Lipids

- Substances derived from simple or compound lipids by hydrolysis.
- **Examples:**
  - **Fatty acids**, **glycerol**, **steroids** (like cholesterol), **vitamin D**, and **hormones** (like estrogen and testosterone)

✅ **Importance / Functions of Lipids:**

Function	Explanation
Energy Storage	Provide more energy than carbohydrates when broken down.
Structural Role	Major components of cell membranes (phospholipids and cholesterol).
Insulation and Protection	Act as a thermal insulator and protect organs like the kidneys and heart.
Hormone Synthesis	Serve as precursors for steroid hormones and vitamin D.
Transport of Fat-Soluble Vitamins	Help absorb and transport vitamins A, D, E, and K.

Waterproofing and Lubrication	Waxes in plants and sebum in skin prevent water loss and reduce friction.

Lipids are essential biomolecules that provide **energy**, form **cell structures**, and support various **biological functions** such as hormone production and insulation.

What is RNA? Mention their type and function.

Ans:

RNA (Ribonucleic Acid) is a **single-stranded nucleic acid** made of **ribose sugar**, **phosphate group**, and **nitrogenous bases** (Adenine, Uracil, Cytosine, Guanine). It plays a crucial role in **protein synthesis** and **gene expression** by carrying instructions from DNA.

Unlike DNA (which is double-stranded and stores genetic information), **RNA acts as a messenger and functional molecule** in the cell.

Types of RNA and Their Functions:

Type of RNA	Full Form	Function
mRNA	Messenger RNA	Carries genetic code from DNA to ribosomes for protein synthesis
tRNA	Transfer RNA	Brings specific amino acids to the ribosome during translation
rRNA	Ribosomal RNA	Structural and functional part of ribosomes where proteins are made
snRNA	Small nuclear RNA	Involved in RNA splicing (removing introns from pre-mRNA)
miRNA	Micro RNA	Regulates gene expression by blocking or degrading mRNA
siRNA	Small interfering RNA	Helps in gene silencing by degrading target mRNA

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information into proteins. Its types—**mRNA, tRNA, rRNA**, and others—work together to ensure that genes are properly expressed and regulated in the cell.

Question: Write about the Anaplerotic role of TCA cycle.

Answer:

Anaplerotic Role of the TCA Cycle – Biochemistry

The **TCA cycle** (Tricarboxylic Acid Cycle), also known as the **Krebs cycle** or **citric acid cycle**, plays a central role in cellular metabolism, not only by producing energy but also by serving an **anaplerotic** function.

What is Anaplerosis?

Anaplerosis refers to the process of **replenishing intermediates** of the TCA cycle that are **withdrawn** for various biosynthetic purposes. These intermediates are used in the synthesis of amino acids, nucleotides, glucose, heme, and fatty acids. Without replenishment, the cycle would slow down or halt, disrupting energy production and biosynthesis.

Anaplerotic Role of the TCA Cycle

The TCA cycle itself **does not produce anaplerotic compounds** but **requires them** to maintain its function. Hence, various **anaplerotic reactions** supply the intermediates to the cycle.

Major Anaplerotic Reactions:

1. **Pyruvate → Oxaloacetate**
    - **Enzyme**: Pyruvate carboxylase (requires biotin)
    - **Location**: Mitochondria
    - **Importance**: Oxaloacetate is crucial for the condensation with acetyl-CoA to begin the cycle. This is the **primary anaplerotic reaction** in most tissues.
2. **Glutamate → α-Ketoglutarate**
    - **Enzyme**: Glutamate dehydrogenase or transaminases
    - **Importance**: Replenishes α-ketoglutarate used in amino acid metabolism.
3. **Aspartate → Oxaloacetate**
    - **Via**: Transamination
    - Supports TCA function and gluconeogenesis.
4. **Propionyl-CoA → Succinyl-CoA**
    - From metabolism of odd-chain fatty acids and some amino acids (e.g., valine, isoleucine, methionine)
    - **Importance**: Replenishes succinyl-CoA, an intermediate of the TCA cycle.
5. **Malate and Fumarate Production**
    - From metabolism of amino acids like phenylalanine and tyrosine.

The **anaplerotic role** of the TCA cycle is crucial for maintaining the balance between **energy production** and **biosynthetic needs**. Since TCA cycle intermediates are siphoned off for biosynthesis, anaplerotic reactions are essential to **replenish these intermediates**, ensuring the continuity and efficiency of the cycle.

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8. How are triglycerides produced?

Ans:

How are Triglycerides Produced?

Triglycerides (also called triacylglycerols) are **esters formed from glycerol and three fatty acids**. They serve as the main form of stored energy in animals and plants.

Steps of Triglyceride Synthesis:

1. **Starting Molecule: Glycerol-3-Phosphate**
   - Glycerol is first phosphorylated or derived from **dihydroxyacetone phosphate (a glycolysis intermediate)** to form glycerol-3-phosphate.
2. **Esterification with Fatty Acids**
   - Three fatty acid molecules, activated as **fatty acyl-CoA**, are sequentially attached to glycerol-3-phosphate.
   - The fatty acids are attached by forming **ester bonds** with the hydroxyl (-OH) groups of glycerol.
3. **Stepwise Formation**
   - First, glycerol-3-phosphate reacts with one fatty acyl-CoA → forms **lysophosphatidic acid**.
   - Second fatty acyl-CoA attaches → forms **phosphatidic acid**.
   - Phosphatidic acid is then dephosphorylated → forms **diacylglycerol (DAG)**.
   - Third fatty acyl-CoA attaches to DAG → forms **triglyceride**.

Overall Reaction:

Glycerol + 3 Fatty Acyl-CoA → Triglyceride + 3 CoA

Where Does This Occur?

- Mainly in the **endoplasmic reticulum of adipose tissue and liver cells**.

What are the factors affecting enzyme activity? Briefly explain any one.

Factors Affecting Enzyme Activity

Enzyme activity can be influenced by several factors that affect the enzyme's shape, function, and the interaction with its substrate. The main factors are:

1. Temperature
2. pH
3. Substrate concentration
4. Enzyme concentration
5. Presence of inhibitors
6. Cofactors and coenzymes
7. Salt concentration (ionic strength

Brief Explanation of One Factor: Temperature

- Enzymes have an **optimal temperature** at which their activity is maximum (usually around 37°C for human enzymes).
- Increasing temperature generally **increases enzyme activity** because molecules move faster, leading to more frequent collisions between enzyme and substrate.
- However, if the temperature rises **beyond the optimum**, the enzyme's **3D structure denatures** (unfolds), causing loss of active site shape and thus **loss of activity**.
- Lower temperatures reduce kinetic energy and enzyme activity but do not typically denature enzymes.

RNA Short note:

RNA (Ribonucleic Acid) is a molecule similar to DNA but with some key differences. It plays a crucial role in the process of using genetic information to make proteins.

- Structure: RNA is usually single-stranded, unlike DNA which is double-stranded.
  Sugar in RNA: Ribose (instead of deoxyribose in DNA)
  Bases: Adenine (A), Uracil (U), Cytosine (C), and Guanine (G)
  Uracil (U) replaces Thymine (T) in RNA
- Types of RNA:
  1. mRNA (Messenger RNA) – carries the genetic code from DNA to ribosomes for protein synthesis.
  2. tRNA (Transfer RNA) – brings amino acids to the ribosome to build proteins.
  3. rRNA (Ribosomal RNA) – forms part of the ribosome and helps in protein synthesis.

- Function: RNA acts as a messenger and helper in the process of making proteins, a process called gene expression.

DNA - Short Note:

DNA (Deoxyribonucleic Acid) is the molecule that carries the genetic instructions used in the growth, development, functioning, and reproduction of all known living organisms and many viruses.

Structure: DNA is shaped like a double helix (a twisted ladder). It is made of units called nucleotides, each consisting of:
- ➢ A phosphate group
- ➢ A sugar (deoxyribose)
- ➢ A nitrogenous base (Adenine [A], Thymine [T], Cytosine [C], or Guanine [G])
- ➢ Base Pairing Rule: A pairs with T, and C pairs with G.

Function: DNA stores genetic information. This information is used to make proteins, which perform most functions in the body.

Location: In eukaryotic cells, DNA is found mainly in the nucleus; in prokaryotic cells, it's found in the cytoplasm.

Lipoprotein – Short Note:

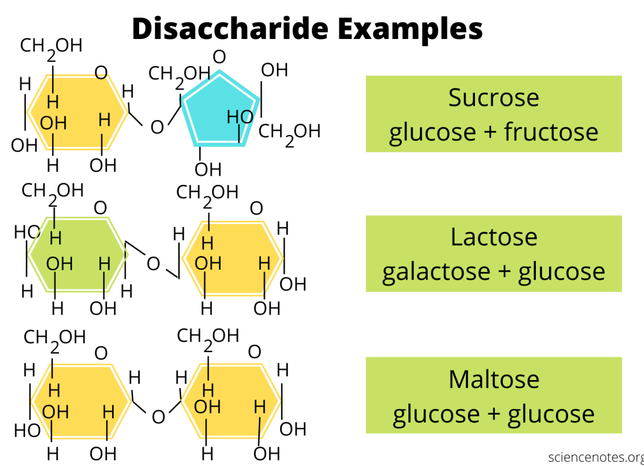
Lipoproteins are molecules made of lipids (fats) and proteins. They help transport fats like cholesterol and triglycerides through the bloodstream, which is mostly water and cannot carry fats alone.

Types of Lipoproteins:

1. HDL (High-Density Lipoprotein) – "Good cholesterol"; removes excess cholesterol from the blood.
2. LDL (Low-Density Lipoprotein) – "Bad cholesterol"; high levels can lead to fat buildup in arteries.
3. VLDL (Very Low-Density Lipoprotein) – carries triglycerides; also contributes to plaque buildup.
4. Chylomicrons – transport dietary lipids from the intestines to other parts of the body.

Function:

Transport lipids in blood
Maintain cholesterol balance
Support cell membrane formation



Question: Classify disaccharide with examples

Ans: Disaccharides are carbohydrates formed by the combination of two monosaccharide units linked by a glycosidic bond.

They are classified based on:

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1. Reducing vs Non-Reducing Disaccharides

A. Reducing Disaccharides:

- Contain a free anomeric carbon that can act as a reducing agent.
- Examples:
  - Maltose = Glucose + Glucose
  - Lactose = Glucose + Galactose

B. Non-Reducing Disaccharides:

- Both anomeric carbons are involved in the glycosidic bond; no free reducing group.
- Example:
  - Sucrose = Glucose + Fructose

Question:write the short note of Central dogma of life.

Ans: The Central Dogma of Life describes the flow of genetic information within a biological system. It was first proposed by Francis Crick in 1958.

DNA → RNA → Protein

Steps in the Central Dogma:

1. DNA → RNA (Transcription)
Genetic information from DNA is transcribed into messenger RNA (mRNA).
Takes place in the nucleus (in eukaryotes).

2. RNA → Protein (Translation)

The mRNA is translated to form proteins (chains of amino acids) with the help of ribosomes and tRNA.
Takes place in the cytoplasm.

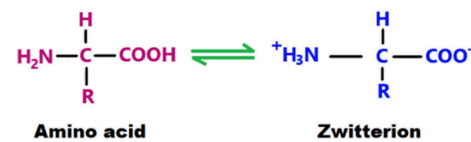
Expanded View (including reverse processes):
Replication: DNA makes copies of itself.
Transcription: DNA → RNA
Translation: RNA → Protein
Reverse Transcription (in some viruses): RNA → DNA (e.g., HIV)

Significance:
This process is essential for gene expression, which ultimately determines the structure and function of all living organisms.

The central dogma of Molecular biology describes the flow of genetic information within a cell, starting from DNA, then transcribed to RNA, and finally translated into proteins. This fundamental principle dictates that genetic information is generally not transferred from proteins to nucleic acids.
Here's a more detailed breakdown:
DNA Replication: DNA can be copied to create more DNA molecules (replication).
Transcription: DNA is used as a template to create RNA molecules (transcription).
Translation: RNA is used as a template to create proteins (translation).

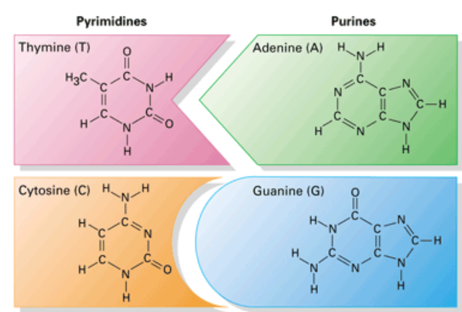
Question: Write briefly about zwitterion.

Ans: zwitterion, also called an inner salt or dipolar ion, is a molecule with both positive and negative charges, but with an overall net charge of zero. It contains an equal number of positively and negatively charged functional groups, meaning it's electrically neutral.



Question: Nitrogenous Base

Nitrogenous bases are organic molecules containing nitrogen that are essential components of DNA and RNA, the building blocks of genetic information. They are classified as either purines (adenine and guanine) or pyrimidines (cytosine, uracil, and thymine). In DNA, the nitrogenous bases are adenine (A), guanine (G), cytosine (C), and thymine (T). In RNA, uracil (U) replaces thymine.



Key points about nitrogenous bases:
Composition: They are made up of carbon, hydrogen, nitrogen, and oxygen atoms, forming ring structures.
Classification: They are divided into purines (two-ring structure) and pyrimidines (single-ring structure).
DNA and RNA: They are crucial for building DNA and RNA, which carry genetic information.
Base Pairing: In DNA, adenine pairs with thymine, and guanine pairs with cytosine. In RNA, adenine pairs with uracil, and guanine pairs with cytosine.
Importance of nitrogenous base: They determine the sequence of nucleotides in DNA and RNA, which in turn determines the genetic information.

Question: Describe the scope of Biochemistry .
Ans:
Biochemistry, a branch of both chemistry and biology, studies the chemical processes within living organisms. Its scope is vast, encompassing various areas from understanding the molecular basis of life to developing new treatments for diseases. It plays a crucial role in diverse fields like medicine, agriculture, food science, and forensics.

Key areas within the scope of biochemistry include:
Medicine and Health Sciences: Understanding biochemical mechanisms of diseases, developing drugs, and creating diagnostic tools.
Agriculture: Improving crop yields, developing pest-resistant crops, and enhancing soil health.
Food Industry: Ensuring food safety and quality, optimizing food processing, and understanding the nutritional value of foods.
Forensic Science: Analyzing trace evidence and identifying individuals.
Molecular Biology and Genetics: Studying the structure and function of DNA and RNA, and understanding inheritance patterns.
Environmental Science: Investigating the effects of pollutants on living organisms and developing sustainable practices.
Industrial Biochemistry: Developing bioproducts and biotechnological processes.
Nutrition: Understanding the biochemical roles of nutrients and their impact on health.
Academic Research: Conducting research to advance our understanding of life's chemical processes.

In essence, biochemistry's scope is as broad as life itself, as it delves into the chemical processes that underpin all living organisms, from microorganisms to humans.

Enlist the biological buffers and write about phosphate buffer.

Ans: Biological buffers are substances that resist changes in pH, playing a crucial role in maintaining homeostasis.

Key biological buffers include bicarbonate, phosphate, proteins, and hemoglobin.

Phosphate buffers, specifically, are essential for regulating pH in intracellular and renal fluids, using dihydrogen phosphate and hydrogen phosphate ions as a weak acid and weak base, respectively.

Phosphate Buffer

Function:
Phosphate buffers help maintain a constant pH in biological fluids, including those within cells and in the kidneys.

Mechanism:
Phosphate buffers utilize two forms of phosphate ions: dihydrogen phosphate ($H_2PO_4^-$) which acts as a weak acid, and hydrogen phosphate (HPO_4^{2-}) which acts as a weak base. When excess hydroxide ions (OH^-) enter the cell, dihydrogen phosphate donates a proton, neutralizing them. Conversely, when excess hydrogen ions (H^+) enter, hydrogen phosphate accepts a proton.
Location:
Phosphate buffers are primarily active in intracellular fluids and in renal tubular fluids.

Importance:
Maintaining a stable pH is crucial for enzyme activity, protein structure, and overall cellular function.

Examples:
Common examples of phosphate buffers include acetic acid/sodium acetate and phosphoric acid/sodium phosphate.

Other Biological Buffers

Bicarbonate Buffer System:
The bicarbonate buffer system is a primary buffer in the extracellular fluid (like blood) and uses carbonic acid (H_2CO_3) and bicarbonate (HCO_3^-).

Protein Buffers:
Proteins, due to their diverse amino acid side chains, can act as buffers, accepting or donating protons to resist pH changes.

Hemoglobin:
Hemoglobin, found in red blood cells, also contributes to buffering, particularly in the transport of carbon dioxide and the regulation of blood pH.

Question: How does prokaryotes differ from eukaryotes?

Ans: Prokaryotic and eukaryotic cells differ primarily in their internal organization and complexity. Prokaryotes lack a membrane-bound nucleus and other membrane-bound organelles, while eukaryotes have both. Prokaryotes are generally smaller and simpler, while eukaryotes are larger and more complex, often involving multicellular organisms.

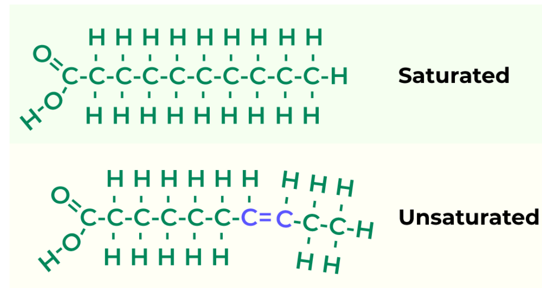
Here's a more detailed comparison:

Prokaryotic Cells	Eukaryotic Cells
Nucleus: Absent. DNA resides in a nucleoid region, not enclosed by a membrane.	Nucleus: Present, containing the cell's genetic material (DNA) enclosed by a nuclear envelope.
Organelles: Lack membrane-bound organelles like mitochondria, endoplasmic reticulum, and Golgi apparatus.	Organelles: Contain numerous membrane-bound organelles, each with specific functions.
Size: Generally smaller, ranging from 0.1 to 5.0 µm in diameter.	Size: Generally larger, ranging from 10 to 100 µm in diameter.
Cell Wall: Many prokaryotes have a cell wall, often composed of peptidoglycan in bacteria .	Cell Wall: Some eukaryotes, like plants, have cell walls, but not all.
Reproduction: Primarily asexual, through binary fission.	Reproduction: Can reproduce both asexually (mitosis) and sexually (meiosis).
DNA Structure: Single, circular chromosome.	DNA Structure: Multiple linear chromosomes, organized within the nucleus.
Examples: Bacteria and Archaea.	

	Examples: Animals, plants, fungi, and protists.

Question: Write the importance of lipid.

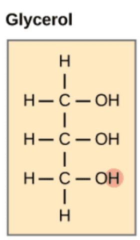
Ans: Lipids are crucial for various bodily functions, serving as energy storage, forming cell membranes, regulating hormones, and insulating organs. They also play a role in nutrient absorption and provide essential fatty acids.



Question: Define Fatty acids.Its structure

Ans: Lipids are fatty compounds that perform a variety of functions in your body. They're part of your cell membranes and help control what goes in and out of your cells. They help with moving and storing energy, absorbing vitamins and making hormones. Having too much of some lipids is harmful.

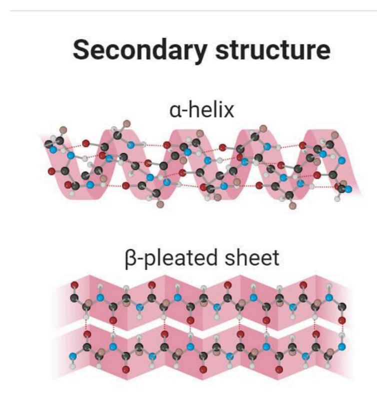
Glycerol



Question: Comment on secondary structure of protein

Ans:

Protein secondary structure refers to the localized folding patterns of a polypeptide chain, primarily involving hydrogen bonds between the backbone atoms. These regular, repeating structures, like alpha-helices and beta-sheets, are stabilized by hydrogen bonds between the carbonyl oxygen and the amide hydrogen of the peptide backbone.



Here's a more detailed look:

Hydrogen Bonds:
The key to secondary structure is the formation of hydrogen bonds between the peptide backbone's carbonyl oxygen and amide hydrogen atoms.

Alpha-helices:
These are coiled structures where the polypeptide backbone forms a right-handed helix, stabilized by hydrogen bonds between the carbonyl oxygen of one amino acid and the amide hydrogen of an amino acid four residues down the chain.

Beta-sheets:
These are formed when several segments of the polypeptide chain lie side-by-side and are connected by hydrogen bonds. They can be parallel or antiparallel.

Other Structures:
Besides alpha-helices and beta-sheets, other less common secondary structural elements include beta-turns (where the polypeptide chain changes direction) and random coils.

Importance:
Secondary structure is crucial for protein folding and function, as it provides the building blocks for the three-dimensional tertiary structure.

Question: Give the outline reaction of glycolysis.

Ans: Glycolysis, in its simplest form, is the breakdown of glucose into pyruvate, producing ATP and NADH.

The overall reaction can be represented as: Glucose + 2 NAD+ + 2 ADP + 2 Pi → 2 Pyruvate + 2 ATP + 2 NADH + 2 H+.

Here's a more detailed breakdown of the process:

1. Energy Investment Phase:

Step 1: Glucose is phosphorylated by hexokinase, using ATP to form glucose-6-phosphate.

Step 2: Glucose-6-phosphate is isomerized to fructose-6-phosphate by phosphoglucoisomerase.

Step 3: Fructose-6-phosphate is phosphorylated by phosphofructokinase, using ATP to form fructose-1,6-bisphosphate.

Step 4: Fructose-1,6-bisphosphate is cleaved by aldolase into two three-carbon molecules: dihydroxyacetone phosphate and glyceraldehyde-3-phosphate.

Step 5: Triose phosphate isomerase converts dihydroxyacetone phosphate into glyceraldehyde-3-phosphate.

2. Energy Payoff Phase:

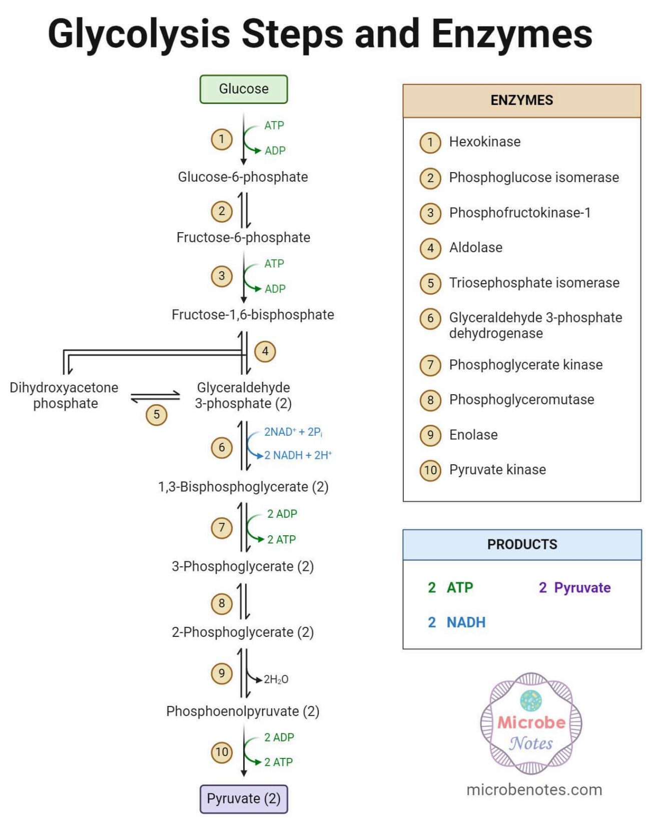
Step 6: Glyceraldehyde-3-phosphate is oxidized by glyceraldehyde-3-phosphate dehydrogenase, reducing NAD+ to NADH and forming 1,3-bisphosphoglycerate.

Step 7: 1,3-bisphosphoglycerate is dephosphorylated by phosphoglycerate kinase, transferring a phosphate to ADP to form ATP and producing 3-phosphoglycerate.

Step 8: 3-phosphoglycerate is converted to 2-phosphoglycerate by phosphoglycerate mutase.

Step 9: 2-phosphoglycerate is dehydrated by enolase to form phosphoenolpyruvate.

Step 10: Phosphoenolpyruvate is dephosphorylated by pyruvate kinase, transferring a phosphate to ADP to form ATP and producing pyruvate.

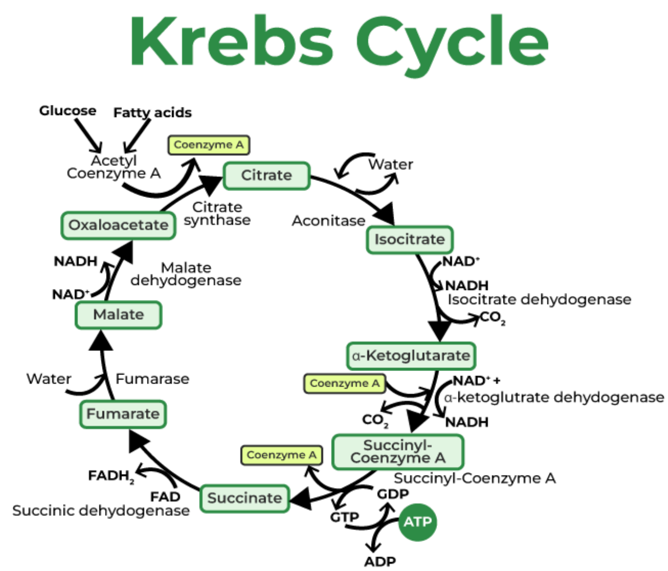


Net Result:

Glycolysis yields a net gain of 2 ATP molecules and 2 NADH molecules per glucose molecule. The pyruvate produced can then proceed to other metabolic pathways, such as the citric acid cycle, or be converted to lactate in anaerobic conditions.

Question: How is the urea cycle linked with TCA cycle?

Ans: The urea cycle and the Tricarboxylic Acid (TCA) cycle, also known as the Krebs cycle or citric acid cycle, are interconnected through the sharing of intermediates like aspartate and fumarate. Fumarate produced in the urea cycle enters the TCA cycle, and aspartate, a TCA cycle intermediate, is involved in the urea cycle.



Elaboration:

1. Aspartate: In the urea cycle, aspartate contributes a nitrogen atom, which is incorporated into urea. Aspartate is also an intermediate in the TCA cycle, produced from transamination of oxaloacetate.
2. Fumarate: The urea cycle produces fumarate, which can be converted to malate and then to oxaloacetate by TCA cycle enzymes.
3. Bicarbonate: The TCA cycle generates CO2, which can be converted to bicarbonate, a precursor for urea synthesis.
4. Oxaloacetate: Oxaloacetate, a TCA cycle intermediate, can be converted to aspartate through transamination, thus connecting the two cycles.
5. NAG: Acetyl CoA, produced by the TCA cycle, can react with glutamate to form N-acetylglutamate (NAG), which regulates the urea cycle.

In essence, the TCA cycle provides intermediates and energy for the urea cycle, and the urea cycle returns metabolites to the TCA cycle. This interconnectedness is often described as a "bicycle" or "shunt," highlighting the dynamic exchange between the two cycles.

Question: What does Km mean in enzyme kinetics? Briefly describe.

Derivation of Michaelis- Menten Equation:

- Following symbol are used for deriving Michaelis- Menten Equation:
  - (E_t) = total concentration of enzymes
  - (S) = total concentration of substrate
  - (ES) = concentration of enzyme-substrate complex
  - $(E_t) - (ES)$ = Concentration of free enzyme

➢The rate of formation of products (i.e., Velocity, V) is proportional to the concentration of the enzyme- substrate complex.

$$V = (ES) \qquad \ldots\ldots(1)$$

➢The maximum reaction rate ,V_m will occur at a point where the total enzyme E_t is bound to the substrate. Then the maximum concentration of ES will be equal to the total enzyme concentration, E_t .

Thus; $V_{max} = (E_t) \qquad \ldots\ldots(2)$

➢Dividing equation (1) by (2);

$$\frac{V}{V_{max}} = \frac{(ES)}{(E_t)} \qquad \ldots\ldots(3)$$

➢Now, for reversible rxn; $ES \rightleftharpoons E + S$

The equilibrium constant for dissociation of ES is given by K_m as;

$$K_m = \frac{(E)(S)}{(ES)}$$

$$K_m = \frac{\{(E_t) - (ES)\}(S)}{(ES)} \qquad \ldots\ldots(4)$$

Or, $(ES)K_m = (E_t)(S) - (ES)(S)$

Or, $(ES)K_m + (ES)(S) = (E_t)(S)$

Or, $(ES)[K_m + (S)] = (E_t)(S)$

Or, $\frac{(ES)}{(E_t)} = \frac{(S)}{K_m + (S)} \qquad \ldots\ldots(5)$

➢Substituting the value of $(ES) / (E_t)$ from equation (3) to (5);

$$\frac{V}{V_{max}} = \frac{(S)}{K_m + (S)}$$

$$\boxed{V = \frac{V_{max}(S)}{K_m + (S)}} \qquad \ldots\ldots(6)$$

➢This is called as Michaelis- Menten Equation

Significance of the Michaelis -Menten equation

- Describes enzymae kinetics
- Determine enzyme efficiency
- Help in drugs inhibitors study
- Determines Vmax and catalytic efficiency
- Efficiency in Metabolic Pathway Regulation

Question: DNA Replication
Ans:
DNA REPLICATION by semi-conservative methods
1. Initation
Origin of Replication in prokaryotes from Orisite
and Eukaryotes have multiple origins on each chromosome.

2. Unwinding DNA Helicase: Helicase unwinds the DNA by breaking H bonds between base pairs.
SSBs prevent the strands from reannealing.
Topoisomerase removes super oils ahead of the replication fork by breaking and rejoining the DNA strand
Primase synthesis a short RNA primer to provide 3-OH for DNA polymerase.

3. Elongation
Leading Strand synthesized continuously in the 5'-3' direction
Lagging Strand synthesized discontinuously as Okazaki fragments
DNA Polymerase I / RNase H remove RNA primers and replace them with DNA.
DNA ligase forms phosphodiester bond sealing Okazaki fragments

4. Termination
In prokaryotes, Tus protein bind to Ter sites to stop Replication
In eukaryotes, telomerase extends telomeres at the chromosome end to prevent loss of genetic material.

Question:Share your idea about acid and base with examples.

Ans: Acids donate hydrogen ions (H^+), while bases accept them. Acids have a sour taste and can corrode metals, like in vinegar (acetic acid) or lemon juice (citric acid). Bases have a bitter taste and feel slippery, like in baking soda (sodium bicarbonate) or soap.
Acids:

- Vinegar (Acetic acid): CH_3COOH.
- Lemon juice (Citric acid): $C_6H_8O_7$.
- Hydrochloric acid (HCl): Found in stomach acid.
- Sulfuric acid (H_2SO_4): Used in car batteries.
- Nitric acid (HNO_3): Used in fertilizers and explosives.
-

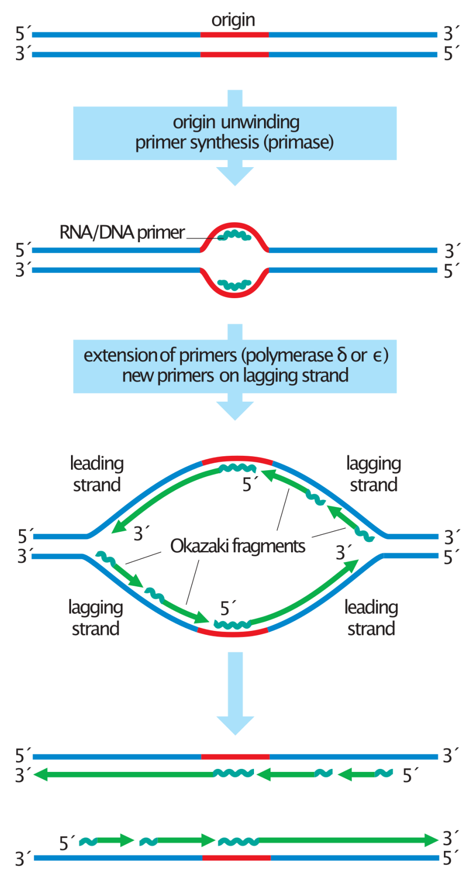
Bases:

- Baking soda (Sodium bicarbonate): $NaHCO_3$.
- Soap: Contains various bases like sodium hydroxide or potassium hydroxide.
- Lime water: Calcium hydroxide ($Ca(OH)_2$).
- Caustic soda (Sodium hydroxide): $NaOH$.
- Milk of magnesia: Magnesium hydroxide ($Mg(OH)_2$).

Question: Write a short note about Polysaccharide

Polysaccharides, or polycarbohydrates, are the most abundant carbohydrates found in food. They are long-chain polymeric carbohydrates composed of monosaccharide units bound together by glycosidic linkages. This carbohydrate can react with water using amylase enzymes as catalyst, which produces constituent sugars. For example: Starch, Glycogen , Cellulose, Chitin, Hyaluronic acid etc

Question: Okazaki fragments



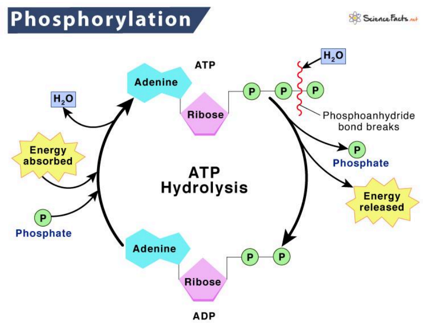
Ans: Okazaki fragments are short, newly synthesized DNA sequences on the lagging strand during DNA replication. They are created in a discontinuous manner, as DNA polymerase can only synthesize DNA in the 5' to 3' direction, but the lagging strand is replicated in the opposite direction. These fragments are later joined together by DNA ligase to form a continuous strand.

Here's a more detailed explanation:
Lagging Strand Replication:
During DNA replication, the lagging strand is copied discontinuously, meaning it's synthesized in short, separate pieces.
Okazaki Fragments:
These short DNA pieces, named after their discoverers Reiji and Tsuneko Okazaki, are the result of this discontinuous synthesis.
RNA Primers:
Each Okazaki fragment is initiated by a short RNA primer.
Joining the Fragments:
After the RNA primers are replaced by DNA and the fragments are correctly positioned, DNA ligase seals the nicks between them, creating a continuous DNA strand.

Importance of Okazaki fragments:
Okazaki fragments are crucial for the replication of the lagging strand and ensure the accurate copying of genetic information.

Phosphorylation

Ans: In biochemistry, phosphorylation is described as the "transfer of a phosphate group" from a donor to an acceptor.
Phosphorylation is a chemical process of adding a phosphate group to an organic compound. Phosphorylation is essential for the functioning of proteins. Because this leads to various activities of modification in several enzymes and in turn regulating their functions.



Question: ETC inhibitors

ETC inhibitors are substances that block the normal flow of electrons through the electron transport chain (ETC), a series of protein complexes in the inner mitochondrial membrane. These complexes are crucial for producing ATP through oxidative phosphorylation. Inhibitors work by binding to specific sites within the ETC complexes, disrupting electron transfer and preventing the creation of a proton gradient. This disruption ultimately leads to a reduction in ATP production.

Electron transport chain complex protein inhibitors are compounds that specifically target and inhibit the activity of one or more of the ETC complexes. These inhibitors are diverse in nature and can act through various mechanisms.

Question: Define PCR.

PCR, or Polymerase Chain Reaction, is a molecular biology technique used to rapidly amplify, or make many copies of, a specific region of DNA. It's often called "molecular photocopying" because it allows scientists to quickly create millions to billions of copies of a DNA sample for analysis.

How it works:

PCR uses a series of temperature changes (denaturation, annealing, and extension) to replicate a specific DNA sequence.

- Denaturation: The double-stranded DNA is heated, causing the strands to separate.
- Annealing: Short DNA sequences called primers bind to specific regions on the separated strands.
- Extension: A DNA polymerase enzyme extends the primers, creating new DNA strands.
- These three steps are repeated multiple times, exponentially increasing the number of copies of the target DNA sequence.

Applications:
PCR has a wide range of applications, including:

- Genetic Testing: Identifying specific DNA sequences for diagnosis of genetic diseases or infectious diseases.
- Forensic Science: Analyzing DNA evidence in criminal investigations.
- Research: Studying gene expression, cloning, and other aspects of molecular biology.
- Medical Diagnostics: Detecting pathogens (disease-causing organisms) in samples like blood, saliva, or tissue.

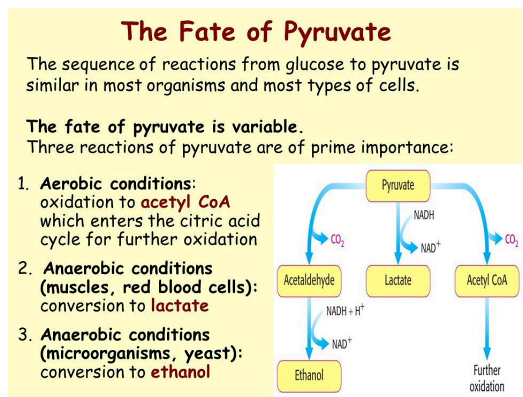
Question: The Fate of Pyruvate

Pyruvate, a product of glycolysis, has several fates depending on cellular conditions and the presence or absence of oxygen. Under aerobic conditions, it's converted to acetyl-CoA, entering the citric acid cycle for energy production. Under anaerobic conditions, it can be reduced to lactate (lactic acid fermentation) or decarboxylated to ethanol (alcohol fermentation). Additionally, pyruvate can be converted to alanine through transamination or to oxaloacetate via pyruvate carboxylase.
Here's a more detailed breakdown:
1. Aerobic Conditions:

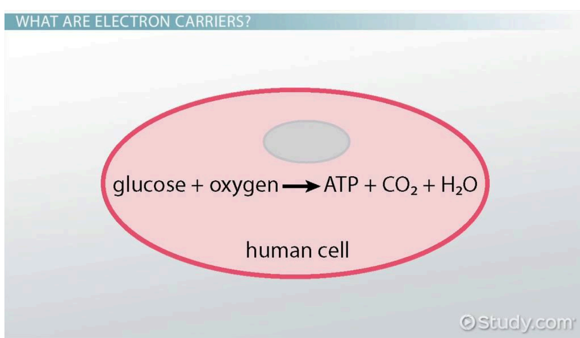
Conversion to Acetyl-CoA:
The pyruvate dehydrogenase complex (PDH) oxidatively decarboxylates pyruvate, converting it into acetyl-CoA. This process is essential for the citric acid cycle, where acetyl-CoA enters to produce ATP.
Oxaloacetate:
Pyruvate can also be converted to oxaloacetate by pyruvate carboxylase, an important step in gluconeogenesis (the synthesis of glucose).

2. Anaerobic Conditions:
Lactic Acid Fermentation:
In the absence of oxygen, pyruvate is reduced to lactate by lactate dehydrogenase, regenerating NAD+ for glycolysis to continue. This process occurs in muscles and some other cells.
Alcohol Fermentation:
In yeast and some other organisms, pyruvate is decarboxylated to acetaldehyde, which is then reduced to ethanol, again regenerating NAD+ for glycolysis.

3. Other Fates:
Alanine:
Pyruvate can be transaminated to alanine, an amino acid, through a reaction catalyzed by alanine aminotransferase. This process is important for muscle metabolism.
Malate:
Pyruvate can be converted to malate by malic enzyme, another route for carbon metabolism.
Acetate:
Pyruvate can also be converted to acetate, either enzymatically or non-enzymatically.



Question: Describe shortly about Electron carrier.



Electron carriers are molecules that transport electrons during cellular respiration and photosynthesis. They accept electrons from one molecule and donate them to another, facilitating the transfer of energy within cells. Examples include NAD+ (which becomes NADH) and FAD (which becomes FADH2).

Key Roles of Electron Carriers:

Energy Transfer:
Electron carriers are essential for energy transfer in cellular respiration and photosynthesis.
Redox Reactions:
They participate in redox reactions, where one molecule is oxidized (loses electrons) and another is reduced (gains electrons).

Electron Transport Chain:
In the electron transport chain (ETC), electron carriers sequentially transfer electrons, releasing energy that is used to create a proton gradient, which ultimately drives ATP production.
Temporary Energy Storage:
Electron carriers temporarily store energy from the oxidation of molecules like glucose.

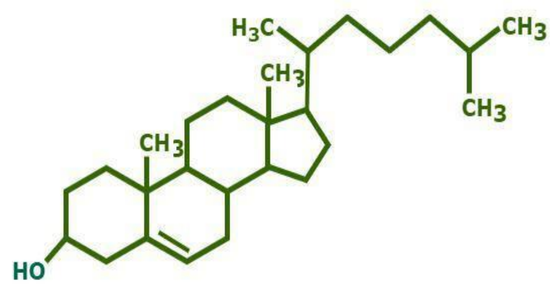
Examples of Electron Carriers:

- **NAD+ (Nicotinamide Adenine Dinucleotide):** NAD+ accepts electrons, becoming NADH in the process.
- **FAD (Flavin Adenine Dinucleotide):** FAD also accepts electrons, becoming FADH2.
- **Coenzyme Q (Ubiquinone):** Coenzyme Q plays a role in the electron transport chain by transferring electrons to complex III.
- **Cytochrome c:** A protein involved in electron transfer, containing a heme group.

Question: What is Cholesterol ? write it's types and structure.

Cholesterol is a waxy, fat-like substance found in all cells of your body. It's essential for building and repairing cells, making hormones, and digesting food. Your body naturally produces all the cholesterol it needs, and it's also found in animal products like egg yolks, meat, and cheese.
Key aspects of cholesterol:
- Essential for bodily functions: Cholesterol is crucial for making cell membranes, hormones, vitamin D, and bile for digestion.
- Naturally produced: Your liver makes all the cholesterol your body needs.
- Found in animal products: You can also get cholesterol from foods like meat, poultry, eggs, and dairy.
- High cholesterol can be a problem: If you have too much cholesterol in your blood, it can lead to plaque buildup in arteries, potentially causing heart disease.

Structure of Cholesterol

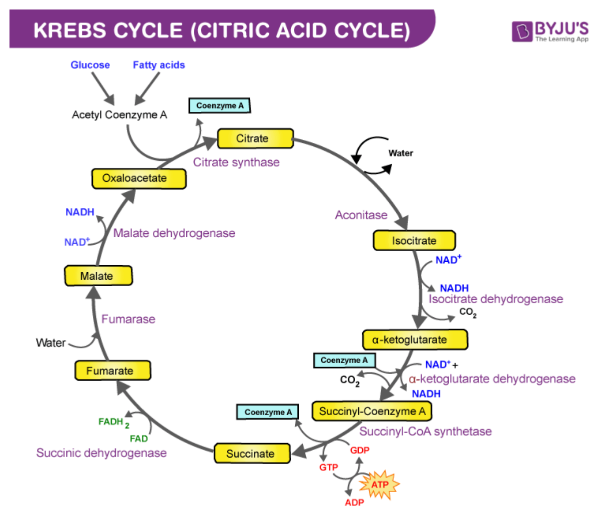


Different types of cholesterol:
- HDL (High-density lipoprotein): Often called "good" cholesterol because it helps your body get rid of excess cholesterol by carrying it back to the liver for removal.
- LDL (Low-density lipoprotein): Often called "bad" cholesterol because it can contribute to plaque buildup in arteries.
- VLDL (Very low-density lipoprotein): Carries triglycerides (another type of fat) in the blood and can also contribute to plaque formation.

Question: Discuss TCA cycle along with the list of the products obtained from this cycle. Explain the significance of Kreb cycle in aerobic condition.

Answer:

The Tricarboxylic Acid (TCA) cycle, also known as the Krebs cycle or citric acid cycle, is a series of metabolic reactions that occur in the mitochondria and generate energy through the oxidation of acetyl-CoA derived from carbohydrates, fats, and proteins. In each cycle, the key products are carbon dioxide, high-energy electron carriers (NADH and FADH2), and GTP/ATP (which can be converted to ATP). The significance of the Krebs cycle in aerobic conditions lies in its role as the central pathway for energy production, generating the electron carriers that power the electron transport chain and ultimately produce ATP.



Products of the TCA Cycle:

For each acetyl-CoA molecule that enters the cycle, the following products are generated:

- **Two molecules of carbon dioxide (CO2):** These are released as waste products.
- **Three molecules of NADH (Reduced Nicotinamide Adenine Dinucleotide):** These are high-energy electron carriers that will donate electrons to the electron transport chain.
- **One molecule of FADH2 (Reduced Flavin Adenine Dinucleotide):** This is also a high-energy electron carrier, also used in the electron transport chain.
- **One molecule of GTP (Guanosine Triphosphate):** This is converted to ATP and is used as an energy currency.

Significance of the Krebs Cycle in Aerobic Conditions:

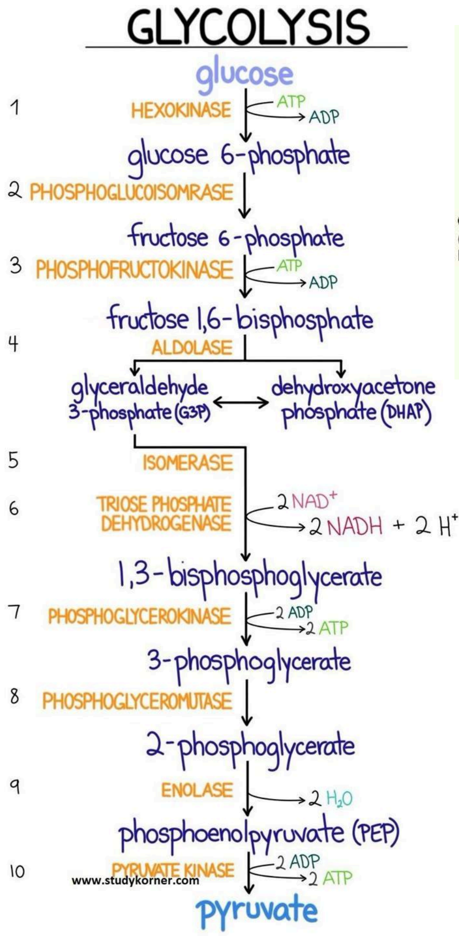
Central Metabolic Pathway:
The TCA cycle is the final common pathway for the complete oxidation of carbohydrates, fats, and proteins.

Energy Production:
The cycle generates high-energy electron carriers (NADH and FADH2) that are then used by the electron transport chain to produce ATP through oxidative phosphorylation.

Metabolic Intermediates:
The cycle also provides intermediate molecules that can be used for the biosynthesis of other molecules like amino acids, fatty acids, and nucleotides.

Regulation of Metabolism:
The TCA cycle is regulated by substrate availability and product inhibition to control energy production and anabolic processes.

Interconnectedness with other Pathways:
The TCA cycle is interconnected with other metabolic pathways, such as glycolysis, fatty acid synthesis, and gluconeogenesis, making it a key hub in cellular metabolism.


GLYCOLYSIS
glucose
1 HEXOKINASE ATP ADP
glucose 6-phosphate
2 PHOSPHOGLUCOISOMRASE
fructose 6-phosphate
3 PHOSPHOFRUCTOKINASE ATP ADP
fructose 1,6-bisphosphate
4 ALDOLASE
glyceraldehyde 3-phosphate (G3P)
dehydroxyacetone phosphate (DHAP)
5 ISOMERASE
6 TRIOSE PHOSPHATE DEHYDROGENASE 2 NAD+ 2 NADH + 2 H+
1,3-bisphosphoglycerate
7 PHOSPHOGLYCEROKINASE 2 ADP 2 ATP
3-phosphoglycerate
8 PHOSPHOGLYCEROMUTASE
2-phosphoglycerate
9 ENOLASE 2 H2O
phosphoenolpyruvate (PEP)
10 PYRUVATE KINASE 2 ADP 2 ATP
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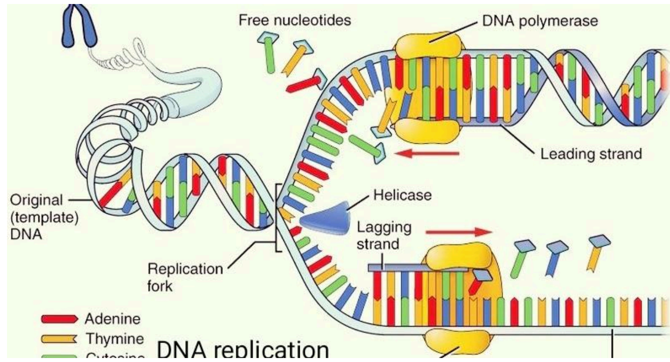
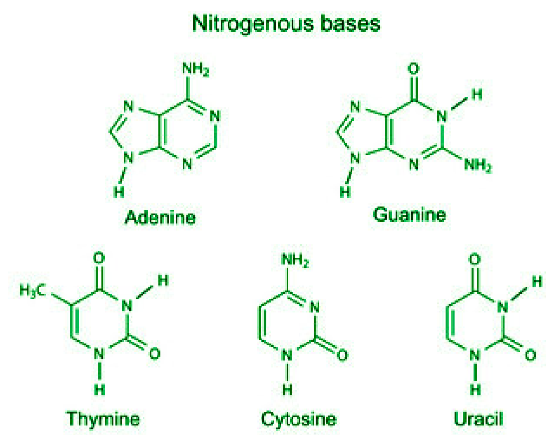
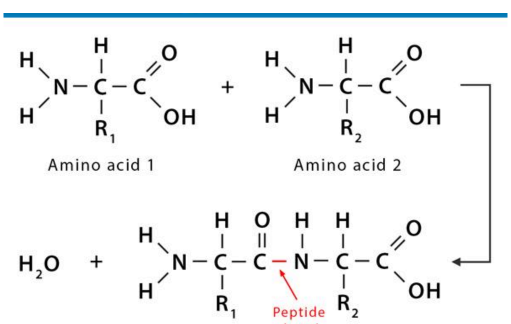
pyruvate


Free nucleotides
DNA polymerase
Leading strand
Original (template) DNA
Helicase
Lagging strand
Replication fork
Adenine
Thymine
DNA replication


Nitrogenous bases
Adenine
Guanine
Thymine
Cytosine
Uracil


Amino acid 1
Amino acid 2
Peptide bond

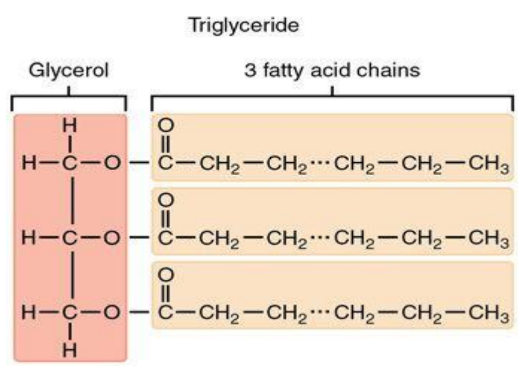
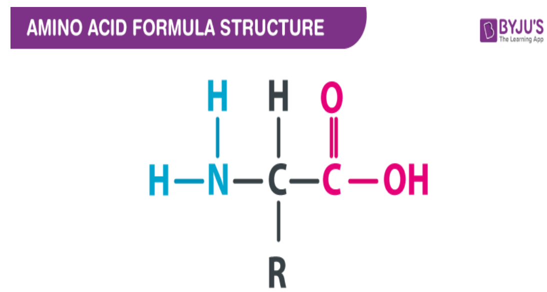
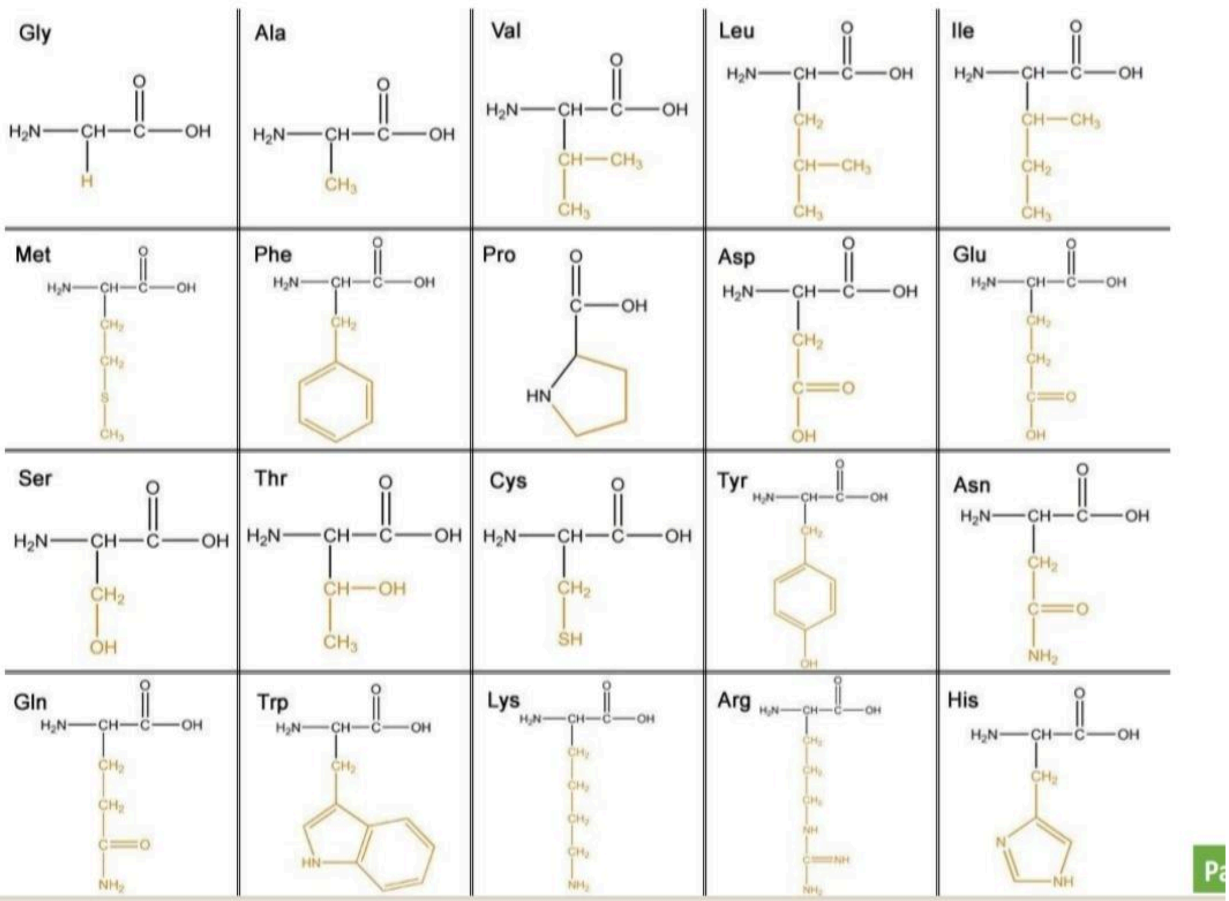

Triglyceride
Glycerol
3 fatty acid chains

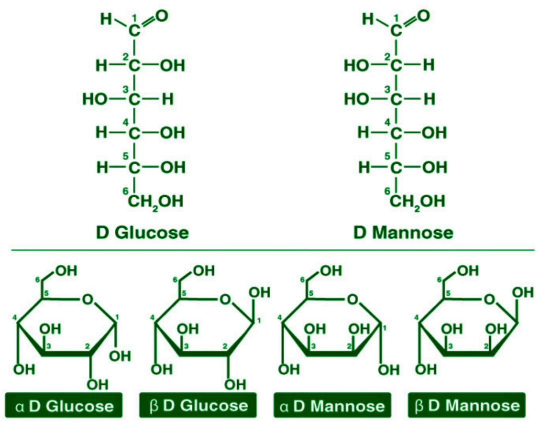
Table: Amino acids & their Symbols

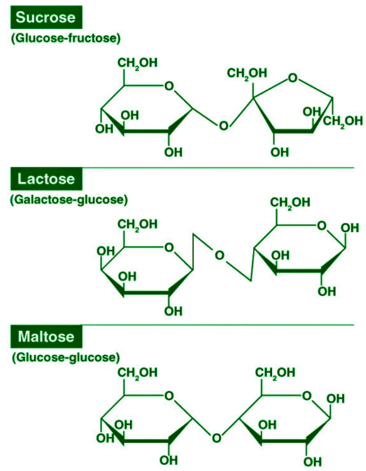
Amino Acids	Symbol	Amino Acids	Symbol
Alanine	Ala	Leucine	Leu
Arginine	Arg	Lysine	Lys
Asparagine	Asn	Methionine	Met
Aspartic acid	Asp	Phenylalanine	Phe
Cysteine	Cys	Proline	Pro
Glutamic acid	Glu	Serine	Ser
Glutamine	Gln	Threonine	Thr
Glycine	Gly	Tryptophan	Trp
Histidine	His	Tyrosine	Tyr
Isoleucine	Ile	Valine	Val

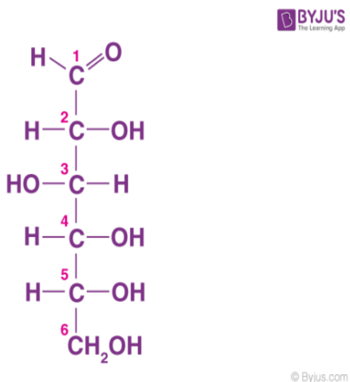


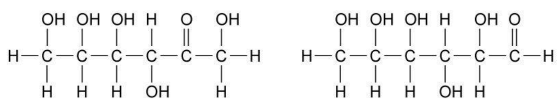
Structures of 20 Standard Amino acids











Fructose Glucose